

FOST TRAINING RULE OF THE ROAD TEST JUL 16

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(G) Vol I and the Mariners Handbook (NP100).

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the OOW training records for inspection by FOST navigation staff. The exam should take no longer than 1 hr.

In this test, any change in the status of the ship or of the environment will be indicated in the script. Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90%.

It is an overcast afternoon and you are the OOW in a T23 (133m). You are currently on passage heading West towards the SCEXA's.

1. You sight ahead a sailing vessel close to your navigation track; you are in open sea. In open water who if anyone shall the sailing vessels keep clear of? (R18b) **(3)**

2. The sailing vessel is now on your R45. She is closing on a steady bearing.

a) Is this a crossing situation within the definition of the Rules? Explain your answer? (R15) **(2)**

b) Who is required to keep out of the way? (R18aiv) **(1)**

c) If required by the Rules, what action shall you take to avoid collision? (R16) **(1)**

Visibility is currently 6nm however restricted vis has been forecast.

3. What constitute 'Restricted visibility' within the meaning of the Rules? (R. 3) **(3)**

The MET forecast is unfortunately accurate. Visibility reduces to 1nm.

4. What are the ranges of your masthead lights, side lights and sternlights? (R22a) **(2)**

5. a) You deem it necessary to sound appropriate signals, what signal do you sound and at what interval? (R35a) **(2)**

b) How long is a 'prolonged blast'? (R32c) **(1)**

6. Through the fog you sight the following dead ahead;

G

W

G

R

a) What type of vessel is this? (R26bi) **(1)**

b) Can you determine, from these lights alone, if the vessel is underway? How? (R26biii) **(1)**

c) What sound signal should this vessel be making? (R35c) **(2)**

- d) What avoiding action should you take? (R16, 18a) (1)

You clear the vessel above and pass out of the fog for the time being.

7. Can you, within the Rules, keep your Navigation lights on? (R20c) (1)

8. Behind you, not visible due to the fog, you hear three blasts in succession; one prolonged blast followed by two short blasts. A vessel engaged in fishing is visual to port and you ascertain the signal is not his.

Other than VEF and Sail, what other types of vessels could this signal indicate? (R35c)
(3)

Thankfully you open from the fog bank and visibility increases to 6nm.

9. You see the vessel astern is a sailing vessel. It is closing on a steady bearing and plot reports the range is reducing.

- a) Who is the give way vessel? Why? (R13) (1)

- b) When and how may you take action to avoid collision? (R17aii) (2)

- c) What sound signal may you sound? (R34d) (1)

The yacht alters on hearing your signal and passes clear down your portside.

10. As they pass you notice a conical shape, apex downwards, in the rigging. What does this indicate? (R25e) (1)

Your reach your RV and the other 4 units involved in the OOW mans form up. The first manoeuvre is a line astern with you at the head. Ahead you sight a merchantman converging with the group. His CPA is in excess of 3nm astern however this will bring him close to the last vessel in the formation.

11. What does ATP 1(G) state about action taken by vessels acting independently or in formation? (ATP1(G) Vol 1) (2)

Following OOW Mans all units are detached and proceed as previously directed. At Green 60, range 4 miles, you sight a power-driven vessel displaying a diamond shape with *two further diamonds visible just above the surface of the water astern of the vessel.*

12. What is this vessel? (R24) (1)

13. What do the two diamonds *astern of the vessel* signify? (R24g) (2)

14. What lights would the vessel display by night? (2)

15. If the bearing of this vessel is steady who is the give way vessel and what action should you take? (R15, R18) (2)

You continue your passage and handover to your relief. Following some rest you return 8 hours later for your evening watch and find the visibility has started to reduce again to about 4 miles and it is starting to get dark with sunset about 20 minutes away.

16. Having taken over the watch your lookout reports sighting a small cluster of lights which looks like a small power driven vessel but a flashing amber light can be seen amongst the navigation lights. What is this likely to be? (1)

17. You are clear of that vessel but notice a small merchant vessel on a similar track to your own that you are slowly overhauling, when shall you be deemed to be overtaking? (R13b) (1)

18. Having cleared the vessel in Q17, you sight the below lights at Green 45 amongst a collection of similar lights.

W
G
W
R
G

a. What is this vessel? (R26 & Annex II) (3)

b. Would any of the vessels lights change if it was not making way through the water? (R26) (1)

19. On completion you continue your passage, and sight ahead the tanker you are to RAS with. The serial proceeds and the guide is ordered to proceed at 10 knots, there is a knot of tidal stream against you, what speed would you expect to be making through the water when in station? (ATP 1G Vol 1) (1)

Your ship takes station on the tanker and passes lines.

- a. What special condition do you now both assume? (R3g) (1)
- b. How would you indicate this by day and by night, if making way? (R27b) (3)

20. You sight ahead HMS OCEAN launching helicopters. After careful observation you calculate that it will pass very close to your RAS group. Who has right of way? (ATP 1G Vol 1) (1)

The RAS finishes and that was your last serial so you continue on your passage bound for Plymouth.

21. While transiting off the Devon coast, you observe smoke and flames coming off a fishing vessel a few miles away. Name 4 other signals can you possibly expect seeing from the vessel in distress? (4)

ANSWERS

It is an overcast afternoon and you are the OOW in a T23 (133m). You are currently on passage heading West towards the SCEXA's.

1. You sight ahead a sailing vessel close to your navigation track; you are in open sea. In open water who if anyone shall the sailing vessels keep clear of? (R18b) (3)

A sailing vessel underway shall keep out of the way of:

- a vessel **Not Under Command**;
- a vessel **Restricted in her Ability to Manoeuvre**;
- a vessel **Engaged in Fishing**.

2. The sailing vessel is now on your R45. She is closing on a steady bearing.

- a) Is this a crossing situation within the definition of the Rules? Explain your answer (R15) (2)

NO - Crossing situation only applies between 2 power driven vessels

(When two *power-driven* vessels are *crossing* so as to involve *Risk of Collision*, the vessel which has the other on her own starboard side shall keep out of the way and shall, if the circumstances of the case admit, avoid *crossing* ahead of the other vessel.)

- b) Who is required to keep out of the way? (R18aiv) (1)

You are;

A power-driven vessel underway shall keep out of the way of:

- a vessel ***Not Under Command***;
- a vessel ***Restricted in her Ability to Manoeuvre***;
- a vessel ***Engaged in Fishing***;
- a sailing vessel.

- c) If required by the Rules, what action shall you take to avoid collision? (R16) (1)

Every vessel which is directed to keep out of the way of another vessel shall, so far as possible, take early and substantial action to keep well clear.

Visibility is currently 6nm however restricted vis has been forecast.

3. What constitutes 'Restricted visibility' within the meaning of the Rules? (R3) (3)

The term '**Restricted Visibility**' means any condition in which visibility is **restricted** by fog, mist, falling snow, heavy rainstorms, sandstorms or any other similar causes.

The MET forecast is unfortunately accurate. Visibility reduces to 1nm.

4. What are the ranges of your masthead lights, side lights and sternlights? (R22a) (2)

In vessels of 50 metres or more in length:

- a masthead light, 6 miles;

- a sidelight, 3 miles;

- a sternlight, 3 miles;

(*1/2 point each with 1/2 point extra if all 3 correct)

5. a) What signal do you sound and at what interval? (R35a) (2)

A power-driven vessel making way through the water shall sound one prolonged blast at intervals of not more than 2 minutes.

- b) How long is a 'prolonged blast'? (R32c) (1)

The term 'prolonged blast' means a blast of four to six seconds' duration.

6. Through the fog you sight the following dead ahead;

G

W

G

R

- a) What type of vessel is this? (R26bi) (1)

A vessel engaged in trawling

- b) Can you determine, from these lights alone, if the vessel is underway? How? (R26biii) (1)

Yes – the display of sidelights indicates he is making way through the water.

- c) What sound signal should this vessel be making? (R. 35c) (2)

A vessel *Engaged in Fishing* shall, instead of the signals prescribed in paragraphs (a) or (b) of this *Rule*, sound at intervals of not more than 2

minutes three blasts in succession, namely one prolonged followed by two short blasts.

d) What avoiding action should you take? (R. 16, 18) (1)

As you are in sight of one another Part B Section II applies and as such a power driven vessel shall keep out of the way of a vessel engaged in fishing. Rule 19 applies to vessels not in sight of one another when navigating in or near an area of Restricted Visibility.

You clear the vessel above and pass out of the fog for the time being.

7. Can you, within the Rules, keep your Navigation lights on? (R. 20c) (1)

Yes - The lights prescribed by these *Rules* shall, if carried, also be exhibited from sunrise to sunset in *Restricted Visibility* and may be exhibited in all other circumstances when it is deemed necessary.

8. Behind you, not visible due to the fog, you hear three blasts in succession; one prolonged blast followed by two short blasts. A vessel engaged in fishing is visual to port and you ascertain the signal is not his.

Other than VEF and Sail what other types of vessels could this signal indicate? (R. 35c) (3)

- a vessel Not Under Command,
 - a vessel Restricted in her Ability to Manoeuvre,
 - a vessel Constrained by her Draught,
 - a sailing vessel,
 - a vessel engaged in Towing or pushing another vessel.
- ($\frac{1}{2}$ point each with additional $\frac{1}{2}$ for naming all 5)

Thankfully you open from the fog bank and visibility increases to 6nm.

9. You see that the vessel astern is a sailing vessel. It is closing on a steady bearing and plot reports the range is reducing.

a) Who is the give way vessel? Why? (R13) (1)

They are - Notwithstanding anything contained in the Rules of Part B, Sections I and II, any vessel overtaking any other shall keep out of the way of the vessel being overtaken.

- b) When and how may you take action to avoid collision? (R17a ii) (2)

The latter vessel may however take action to avoid collision by her manoeuvre alone, as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action in compliance with these Rules.

- c) What sound signal may you sound? (R34d) (1)

When vessels in sight of one another are approaching each other and from any cause either vessel fails to understand the intentions or actions of the other, or is in doubt whether sufficient action is being taken by the other to avoid collision, the vessel in doubt shall immediately indicate such doubt by giving at least five short and rapid blasts on the whistle. Such signal may be supplemented by a light signal of at least five short and rapid flashes.

The yacht alters on hearing you signal and passes clear down your portside.

10. As they pass you notice a conical shape, apex downwards, in the rigging. What does this indicate? (R25e) (1)

A vessel proceeding under sail when also being propelled by machinery shall exhibit forward, where it can best be seen, a conical shape, apex downwards.

Your reach your RV and the other 4 units involved in the OOW mans form up. The first manoeuvre is a line astern with you at the head. Ahead you sight a merchantman converging with the group. His CPA is in excess of 3nm astern however this will bring him close to the last vessel in the formation.

11. What does ATP 1(G) state about action taken by vessels acting independently or in formation? (ATP1(G) Vol 1 Art 2241) (2)

Individual Action to Avoid Danger (ATP 1(G) Vol 1 Art 2241):

Ships, whether acting independently or in formation, must take such individual action as may be required to avoid danger. When necessary, ships may leave their stations to avoid Risk of Collision or to avoid navigational hazards.

On completion of that serial, the Ship continues independently for the next evolution and you sight, at Green 60, range 4 miles, a power-driven vessel displaying a diamond shape with *two further diamonds visible just above the surface of the water astern of the vessel.*

12. What is this vessel? (R24) (1)

Vessel engaged in towing operations

13. What do the two diamonds *astern of the vessel* signify? (R.24g) (2)

An inconspicuous, partly submerged vessel or object, or combination of such vessels or objects where the length of tow is over 200 metres.

(A diamond shape at or near the aftermost extremity of the last vessel or object being towed and if the length of the tow exceeds 200 metres an additional diamond shape where it can best be seen and located as far forwards as is practicable.)

14. What lights would the vessel display by night? (2)

As length of tow is over 200m the towing vessel will display three masthead lights in a vertical line, side lights, a towing light and stern light.

15. If the bearing of this vessel is steady who is the give way vessel and what action should you take? (R15, R18) (2)

Despite the fact the vessel is towing it is not indicating restricted in her ability manoeuvre and therefore, as the vessel is on your starboard bow, you are the give way vessel in accordance with Rule 15 (crossing vessels).

You continue your passage and handover to your relief. Following some rest you return 8 hours later for your evening watch and find the visibility has started to reduce again to about 4 miles and it is starting to get dark with sunset about 20 minutes away.

16. Having taken over the watch your lookout reports sighting a small cluster of lights which looks like a small power driven vessel but a flashing amber light can be seen amongst the navigation lights. What is this likely to be? (1)

Submarine navigating on the surface.

17. You are clear of that vessel but notice a small merchant vessel on a similar track to your own that you are slowly overhauling, when shall you be deemed to be overtaking? (R. 13b) (1)

A vessel shall be deemed to be overtaking when coming up with another vessel from a direction more than 22.5 degrees abaft her beam, that is, in such a position with reference to the vessel she is Overtaking, that at night she would be able to see only the sternlight of that vessel but neither of her sidelights.

18. Having cleared the vessel in Q17, you sight the below lights at Green 45 amongst a collection of similar lights.

W

G

W
R

W
G

- a. What is this vessel? (R26 & Annex II) (3)

Vessel engaged in trawling, probably over 50 metres in length, underway and making way, hauling nets.

- b. Would any of the vessels lights change if it was not making way through the water? (R26) (1)

The vessel only shows sidelights when making way through the water.

19. On completion you continue your passage, and sight ahead the tanker you are to RAS with. The serial proceeds and the guide is ordered to proceed at 10 knots, there is a knot of tidal stream against you, what speed would you expect to be making through the water when in station? (ATP 1G Vol 1) (1)

Para 2120 Course and Speed. Signalled speed is attained by using the normal number of revolutions (or lever settings) for the ordered speed. Therefore speed through the water should be 10 knots regardless of tidal stream.

Your ship takes station on the tanker and passes lines.

- a. What special condition do you now both assume? (R3g) (1)

A vessel restricted in her ability to manoeuvre.

- b. How would you indicate this by day and by night, if making way? (R27b) (3)

- By day three shapes in a vertical line where they can best be seen. The highest and lowest of these shapes shall be balls and the middle one a diamond.

- By night three all-round lights in a vertical line where they can best be seen. The highest and lowest of these lights shall be red and the middle light shall be white.

- In addition, when making way through the water, masthead lights, sidelights and a sternlight.

20. You sight ahead HMS OCEAN launching helicopters. After careful observation you calculate that it will pass very close to your RAS group. Who has right of way? (ATP 1G Vol 1) (1)

Ships engaged in replenishment (other than VERTREP) have the right of way over carriers and other ships engaged in flight operations.

The RAS finishes and that was your last serial so you continue on your passage bound for Plymouth.

21. While transiting off the Devon coast, you observe smoke and flames coming off a fishing vessel a few miles away. Name 4 other signals can you possibly expect seeing from the vessel in distress? (4)

- Rockets or shells, throwing red stars fired one at a time at short intervals;
- International Code Signal of distress indicated by N.C.;
- Signal consisting of a square flag having above or below it a ball or anything resembling a ball;
- Rocket parachute flare or hand-flare showing a red light;
- Smoke signal giving off orange-coloured smoke;
- Slowly and repeatedly raising and lowering arms outstretched to each side.

Total _____ /54